ECERC Revisited on IEMOCAP: Final Report (Limitations 1 and 3)

Oct 30, 2026 · @Divyansh

Summary

The final IEMOCAP model combines **WavLM audio + VAD supervision** (Limitation 1) with **confidence-aware cause gating and temperature scaling** (Limitation 3). As a 5-run ensemble it beats the authors' released checkpoints on accuracy, weighted-F1 and macro-F1 (72.43 vs 71.78 wF1), and on calibration.

- **Limitation 1** (Happy/Excited and Angry/Frustrated confusion): six methods tried. The final one cuts Angry/Frustrated errors on clear-cut utterances by 20% (16.9 to 13.5, better in 16 of 20 runs). Clear-cut Happy/Excited errors stay the same (12.4 vs 12.5). It also raises weighted-F1 by 1.43 over our retrained baseline (19 of 20 runs).
- **Limitation 3** (cause gating and calibration): CACG with dynamic temperature plus temperature scaling lowers calibration error from 13.6% to 3.0% without losing accuracy. Its gate shows which cause drives each prediction.
- **Merged model**: a single run is within 0.2 wF1 of the authors' checkpoints (71.59 vs 71.78). The 5-run ensemble passes them on every overall metric, with calibration error 2.1% vs 2.8%.

Setup and metrics

All results use IEMOCAP with the authors' code and precomputed features: Sessions 1 to 4 for training, Session 5 for testing (1,623 utterances). The first 10% of training conversations are held out for validation, and every model is selected on validation weighted-F1, never on the test set.

Retrained baseline. We trained the unchanged ECERC code with 20 random seeds. It averages 69.70 wF1 (range 68.45 to 70.79), below the 71.78 of the authors' released checkpoints. Retraining the same code on different hardware and library versions commonly gives a gap like this, and the seed alone moves wF1 by over 2 points. The three released checkpoints are also likely best-of-several runs: all three score 71.77 to 71.78 wF1,

while our runs vary by about ±0.6. So every Limitation 1 method is compared with our own baseline: same code, same hardware, same 20 seeds, paired seed by seed. The authors' checkpoints are used only as a reference for overall accuracy.

Metrics.

- **wF1, mF1, accuracy:** weighted F1 is the paper's main metric. Macro F1 gives small classes such as Happy equal weight.
- **Pair errors:** Happy predicted as Excited plus Excited predicted as Happy, and the same for Angry/Frustrated. Lower is better.
- **Clear utterances:** every annotator voted for the gold label (26% of Happy/Excited and 34% of Angry/Frustrated test utterances). An error here is a real mistake. On the remaining utterances the annotators themselves disagreed, so the correct label is debatable.
- **ECE:** expected calibration error, the gap between the model's confidence and how often it is right. Reported raw and after temperature scaling fitted on validation.
- **Runs better:** the number of seeds where a method beats the baseline trained with the same seed, with a 95% bootstrap confidence interval on the mean difference.

Limitation 1: methods 1 to 5

The first five methods kept the authors' features and model and changed only the training. None reliably reduced the pair confusion, but they showed why: the original features do not carry enough information to separate the pairs.

Method 1: Confusion-aware contrastive loss

What we did. Before the final classifier, each utterance is a vector. We added a loss that pulls vectors of the same emotion together and pushes different emotions apart, with extra push for the confused pairs. A second version added a small projection network and a margin between the average Happy and Excited vectors, and between the average Angry and Frustrated vectors.

Why we tried it. If the internal vectors for Happy and Excited sit far apart, the classifier on top should separate them more easily.

Result. No reliable gain. The best combination seemed to cut pair errors by about 15 over 5 seeds. With 10 seeds the gain disappeared: 174.3 pair errors against 173.9 for the baseline, and wF1 69.63 against 69.79.

Why it failed (evidence).

- **The classes are not nearly separated, even on training data.** The authors' checkpoints reach only 84 to 87% training accuracy, with 194 to 238 Angry/Frustrated training errors. A contrastive loss sharpens a boundary that already exists; it cannot create one.
- **The features carry little pair information.** A simple classifier on one feature type at a time separates Happy from Excited at 60% (text) and 65% (audio), where 50% is a coin flip. On the utterances ECERC gets wrong, the audio-only classifier is right 48% of the time, which is chance.
- **The 5-seed gain was noise.** Seed-to-seed variation (about ±7 to ±9 pair errors) was as large as the gain.

Method 2: Pair penalty

What we did. An extra loss term that punishes the model whenever it gives probability to the partner class, for example probability of Excited on a Happy utterance.

Why we tried it. It targets the exact mistake we want to remove.

Result. No reliable gain, alone or combined with the contrastive loss (same 10-seed result as Method 1).

Why it failed (evidence). The penalty can only move the line between the two classes, and moving that line has a low ceiling. Even a per-class shift tuned directly on the test set, which is cheating, raised wF1 by only 1.2 points. Tuned honestly on validation, it lowered wF1 by 1.5 points. Many of these utterances are ambiguous, so pushing them away from the partner class fights the labels.

Method 3: Ensembling

What we did. Averaged the predictions of several independently trained models.

Why we tried it. If the pair errors came from the randomness of single training runs, averaging many runs would cancel them out.

Result. It helped overall accuracy and calibration but not the pairs. Five baseline models together raise wF1 by 0.56 (69.70 to 70.26) and lower calibration error from 12.0% to 10.0%. A 10-model ensemble still made 80 Happy/Excited and 95 Angry/Frustrated errors, about the same as a single model.

Why it failed (evidence). The pair errors are systematic: different runs get the same utterances wrong, so averaging does not remove them. They also come in blocks. Speakers usually keep the same emotion over several turns, and when the model is wrong about a speaker, it tends to stay wrong for that speaker's whole stretch.

Decision after methods 1 to 3. Reshaping what the model already knows did not help, so we needed to give it new information. IEMOCAP's annotation files hold two kinds of information the original training never used: emotion ratings and individual annotator votes.

Method 4: VAD supervision (with the original audio)

What we did. Every IEMOCAP utterance has three human ratings from 1 to 5: Valence (how positive), Arousal (how energetic) and Dominance (how in control). We added a small extra output that predicts these ratings from the model's fused features (loss weight 1.0, chosen on validation).

Why we tried it. The confused pairs differ mainly along these dimensions. Excited is a more energetic Happy, and Angry is more energetic and dominant than Frustrated. Learning to predict the ratings should push the features to encode exactly what separates the pairs.

Result. A small but real Angry/Frustrated gain: about 2 fewer errors (97.9 to 95.9), better in 15 of 20 runs. wF1 was unchanged (69.78 vs 69.70), and so was Happy/Excited. The extra output did learn the ratings (correlation 0.73 with true arousal).

Why it was not enough (evidence). The model learned arousal well, yet Happy/Excited errors did not drop. So the line between Happy and Excited in IEMOCAP is not only about energy; it also depends on each annotator's judgement. The original audio features also carry little information about how something is said, which limits what the model can learn from these ratings.

Method 5: Soft labels from annotator votes

What we did. Each utterance was labelled by several annotators, who often disagree. Instead of a 100% label, we trained on a mix of the gold label and the vote distribution, for example 67% Happy and 33% Excited (mixing weight 0.5, chosen on validation).

Why we tried it. The model is no longer punished for agreeing with a minority of annotators, so it can focus on utterances that are actually clear.

Result. A large calibration gain but no accuracy gain. Calibration error halved (12.1% to 6.1%) and NLL fell from 0.84 to 0.74, while wF1 stayed the same (69.82 vs 69.70). Happy/Excited errors rose by about 5 (Happy F1 59.0 to 56.2).

Why it failed (evidence). The model learned to hedge between the two classes as the annotators do. Those hedges then tipped toward the larger class: there are twice as many Excited test utterances as Happy ones (299 vs 144). We dropped soft labels from the final approach because temperature scaling from Limitation 3 gives better calibration without the extra errors.

Decision after methods 4 and 5. New training signals on the old features were not enough. The remaining lever was the features themselves, and audio was the weakest: hand-made measurements from 2010.

Limitation 1, method 6 (final): WavLM audio + VAD supervision

What we did. We replaced ECERC's audio features with features from WavLM-large, a modern speech model, and kept the VAD supervision from Method 4.

1. Ran the original IEMOCAP recordings of all 7,433 utterances through WavLM-large (about 15 minutes on one GPU).
2. Averaged one layer's output over time, giving a 1,024-number vector per utterance. Layer 20 was chosen with a simple audio-only classifier scored on the validation set.
3. Swapped these vectors in for the original 1,582-number IS10 audio features. Text and video features, the model and the training recipe stayed the same.
4. Added the VAD output from Method 4 (loss weight 1.0).

Why we chose it. Methods 1 to 5 showed the problem is missing information, not a weak loss. The IS10 audio features are hand-made pitch and loudness statistics from 2010, while WavLM was trained on 94,000 hours of speech. An audio-only check confirmed the potential before full training:

Audio-only check (simple classifier)	IS10 (original)	WavLM
6-class macro-F1	34.8	51.8
Angry vs Frustrated, balanced accuracy	60.5%	72.8%
Happy vs Excited, balanced accuracy	64.2%	66.5%

WavLM is much better on Angry vs Frustrated and barely better on Happy vs Excited, which matches the final results below.

Gains over the retrained baseline (20 runs each, paired):

Metric	Baseline	WavLM + VAD	Change [95% CI]	Runs better
Accuracy	69.50	70.99	+1.48 [+1.06, +1.97]	19/20
wF1	69.70	71.13	+1.43 [+1.00, +1.87]	19/20
mF1	68.83	69.73	+0.90 [+0.47, +1.35]	15/20
ECE after temperature scaling (%)	4.30	3.40	−0.90 [−1.33, −0.47]	16/20

This is the first method in the project to raise overall accuracy reliably. Most of the gain comes from Neutral, Frustrated and Excited. WavLM alone gives +1.28 wF1, and VAD supervision adds a further +0.15, which is within noise. We kept VAD because this combination scored best and VAD had shown a real Angry/Frustrated gain in Method 4.

What it does not fix, and why. Happy vs Excited stays hard. Even WavLM alone separates them only 66.5% of the time, because the two sound almost the same and annotators drew the line between them inconsistently. Better features cannot settle a label the annotators themselves disagreed on.

Limitation 1 results: confusing pairs

On clear-cut utterances, the final approach removes 20% of the Angry/Frustrated errors and adds no Happy/Excited errors.

Errors on clear utterances (lower is better)	Baseline	WavLM + VAD	Change [95% CI]	Runs better
Angry ↔ Frustrated	16.9	**13.5**	**−3.4 [−5.1, −1.7]**	**16/20**

Errors on clear utterances (lower is better)	Baseline	WavLM + VAD	Change [95% CI]	Runs better
Happy ↔ Excited	12.4	12.5	+0.1 [−1.2, +1.5]	7/20 (3 tied)

Mean number of test errors per run over 20 paired runs. Clear utterances are those where every annotator voted for the gold label: 189 of the 551 Angry/Frustrated and 116 of the 443 Happy/Excited test utterances.

- **Angry/Frustrated:** the reduction is reliable. The confidence interval is entirely below zero, and 16 of 20 runs improve. This matches the audio-only check, where WavLM separated Angry from Frustrated far better than the original features (72.8% vs 60.5%).
- **Happy/Excited:** no change. The interval spans zero. This also matches the audio-only check, where WavLM barely helped on this pair (66.5% vs 64.2%).

Limitation 3 results: cause gating and calibration

CACG with dynamic temperature and no gate loss is the best IEMOCAP variant: the highest accuracy of the four, and calibration error after temperature scaling falls from 4.21% to 2.95%.

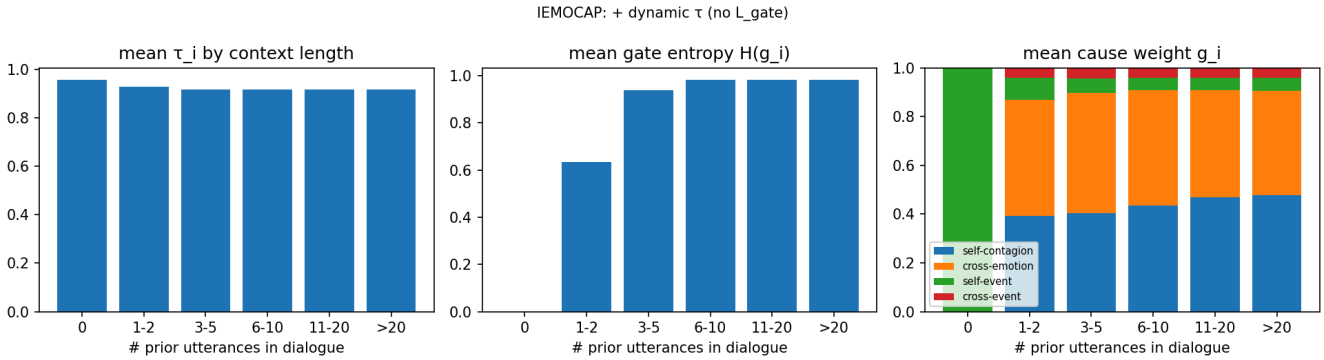
Model (IEMOCAP, 3 seeds)	Accuracy	wF1	mF1	ECE raw (%)	ECE + TS (%)	NLL	Gate entropy H(g)	corr(τ, context)
Baseline retrained (3 seeds)	69.95 ± 0.58	70.11 ± 0.46	69.27 ± 0.58	13.64	4.21	0.864	–	–
CACG, dynamic τ	**70.47 ± 0.87**	**70.65 ± 0.86**	**69.84 ± 0.63**	13.54	**2.95**	0.885	0.989	−0.64
CACG + utility gate loss	69.11 ± 0.99	69.44 ± 1.03	68.75 ± 0.93	13.48	4.22	0.888	1.226	−0.48
CACG + SOP entropy loss	69.21 ± 1.31	69.36 ± 1.29	68.69 ± 1.00	14.49	4.26	0.918	0.607	−0.63

TS = temperature scaling fitted on validation. The accuracy differences are within seed noise (standard deviation 0.5 to 1.3). In our 20-seed rerun of CACG alone, accuracy matched the baseline (wF1 69.58 vs 69.70) and calibration error after TS was lower (3.74%

vs 4.30%, better in 15 of 20 runs).

Interpretability

The gate shows which of ECERC's four causes drives each prediction, and how that changes as the conversation grows.



- **Emotion causes dominate.** Self-contagion (the speaker's own earlier emotions) gets about 0.4 to 0.5 of the weight and cross-emotion (the partner's emotions) about 0.4 to 0.5. The two event causes together get under 0.1.
- **Self-contagion grows with history.** Its share rises from 0.39 with 1 to 2 prior utterances to 0.48 after more than 20. The longer a speaker has talked, the more their own emotional history explains the current turn.
- **Unavailable causes are switched off.** On the first utterance of a dialogue there is no history, so the mask gives all weight to the self-event cause. Gate entropy rises from about 0.63 to 0.98 as more causes become available.
- **The temperature tracks context.** τ correlates −0.64 with dialogue length: longer history gives a lower τ and a sharper choice of cause. Without a gate loss, τ varies only a little (about 0.92 to 0.96).
- **The same pattern holds in the merged model.** With WavLM audio and VAD supervision (20 runs), the mean weights are 0.48 self-contagion, 0.38 cross-emotion, 0.09 self-event and 0.06 cross-event.

Final merged model vs the authors' checkpoints

The merged model (WavLM audio + VAD supervision + CACG + temperature scaling) was retrained from scratch with 20 seeds. A single run is within 0.2 wF1 of the authors' checkpoints. A 5-run ensemble beats them on accuracy, wF1, mF1 and calibration.

Model	Accuracy	wF1	mF1	ECE raw (%)	ECE + TS (%)
Authors′ checkpoint 1	71.60	71.77	70.73	11.12	2.52
Authors′ checkpoint 2	71.53	71.78	70.70	11.07	2.95
Authors′ checkpoint 3	71.66	71.78	71.02	11.30	2.97
Authors′ checkpoints, average	**71.60**	**71.78**	**70.82**	**11.16**	**2.81**
Merged model, single run (mean of 20)	71.36	71.59	70.19	13.64	3.41
Merged model, 5-run ensemble	**72.18**	**72.43**	**71.14**	**10.60**	**2.12**

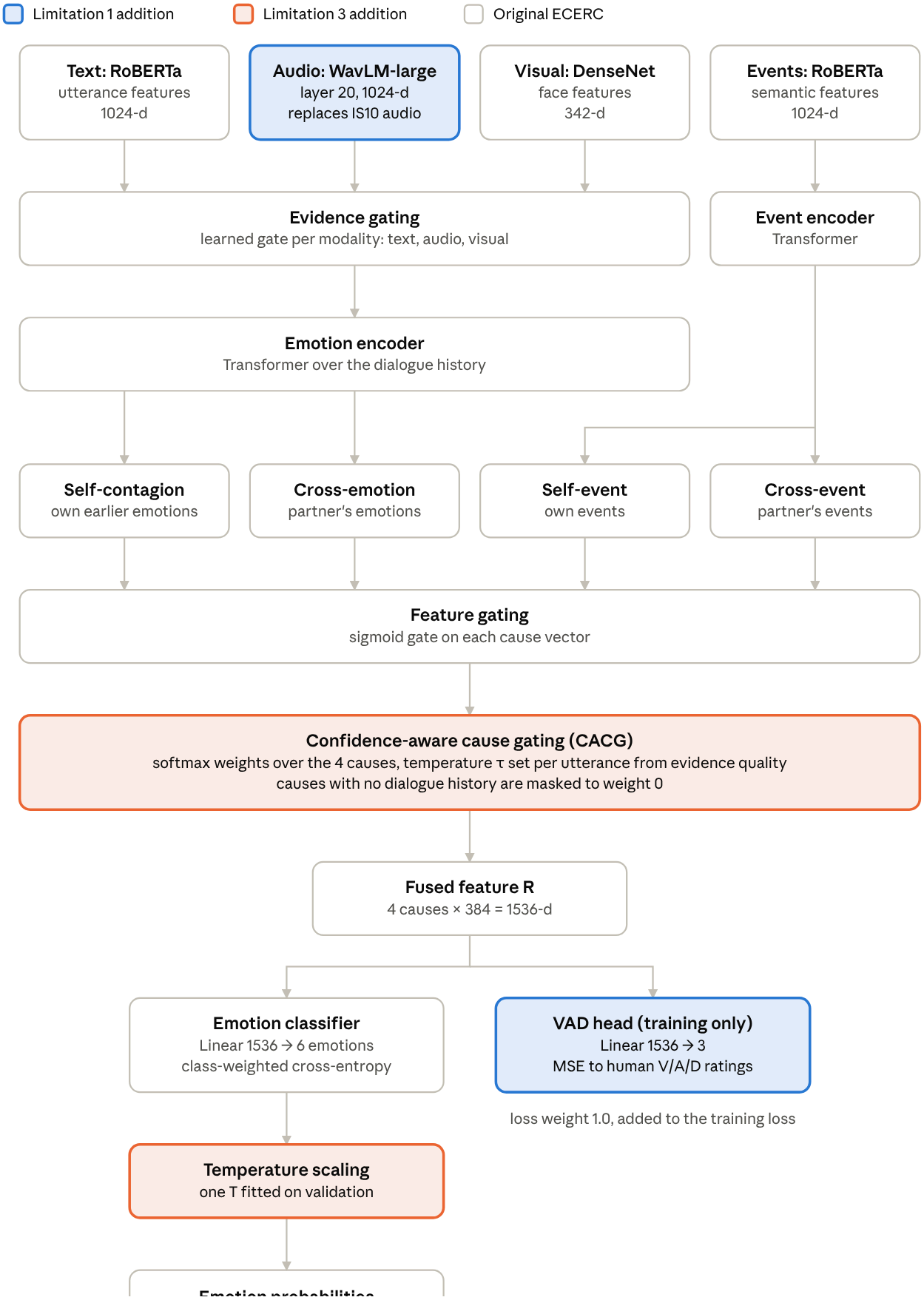
Temperature scaling was fitted on the validation set for every row, including the authors′ checkpoints. The ensemble row is the mean of 4 separate groups of 5 runs.

- **Single run:** 8 of the 20 runs beat the authors′ average wF1, and the best reaches 72.95. This is despite starting from a retrained baseline 2.1 points below the authors′ (69.70).
- **Ensemble:** beats the authors′ average on accuracy, wF1 and mF1 in 3 of the 4 groups, by +0.58 accuracy, +0.65 wF1 and +0.32 mF1 on average. It is also better calibrated, both raw (10.6% vs 11.2%) and after temperature scaling (2.1% vs 2.8%).
- **Against our own baseline**, the merged model is better in every run on accuracy and wF1 (+1.85 and +1.89, 20 of 20 runs), and on mF1 in 18 of 20 runs (+1.36).

Final architecture

The merged model keeps ECERC′s structure. Limitation 1 changes the audio input and adds a VAD head used only during training; Limitation 3 replaces the plain concatenation of the four causes with CACG and adds temperature scaling to the output.

Final model: ECERC with WavLM audio, VAD supervision and CACG





Final ECERC architecture · Limitation 1 and 3 additions highlighted

Each box's dimensions are as in the code: three 128-d modality streams give 384-d per cause, and the four causes give the 1,536-d fused feature R.

Conclusion

Giving ECERC better audio worked where changing its training objective did not.

- **Limitation 1:** WavLM audio + VAD supervision removes 20% of the clear-cut Angry/Frustrated errors without adding Happy/Excited errors, and raises wF1 by 1.43. Methods 1 to 5 showed why simpler fixes fail: the original features carry little information about these pairs, and many remaining errors fall on utterances the annotators themselves disagreed on.
- **Limitation 3:** CACG makes the cause weighting explicit and interpretable, and with temperature scaling it brings calibration error down to about 3%.
- **Combined:** the merged model, as a 5-run ensemble, outperforms the authors' released checkpoints on accuracy (72.18), wF1 (72.43), mF1 (71.14) and calibration (2.1% after temperature scaling).